



Machine Learning

AIML CZG565

M8 : Bayesian Learning

BITS Pilani

Pilani Campus

Course Faculty of M.Tech Cluster
BITS – CSIS - WILP

Disclaimer and Acknowledgement



- These content of modules & context under topics are planned by the course owner Dr. Sugata, with grateful acknowledgement to many others who made their course materials freely available online
- We here by acknowledge all the contributors for their material and inputs.
- We have provided source information wherever necessary
- Students are requested to refer to the textbook w.r.t detailed content of the presentation deck shared over canvas
- We have reduced the slides from canvas and modified the content flow to suit the requirements of the course and for ease of class presentation

Slide Source / Preparation / Review:

From BITS Pilani WILP: Prof.Sugata, Prof.Chetana, , Prof.Monali, Prof.Rajavadhana

External: CS109 and CS229 Stanford lecture notes, Dr.Andrew NG and many others who made their course materials freely available online

Course Plan

M1	Introduction & Mathematical Preliminaries
M2	Machine Learning Workflow
M3	Linear Models for Regression
M4	Linear Models for Classification
M5	Decision Tree
M6	Instance Based Learning
M7	Support Vector Machine – Planned to be discussed in CS 15 or CS 16
M8	Bayesian Learning
M9	Ensemble Learning
M10	Unsupervised Learning
M11	Machine Learning Model Evaluation/Comparison

Par. Est = MLE
all h are
eq. likely



Maximum A Posteriori (MAP) Analysis

100 training eq.
+ve -ve

1. Determine prior probability

2. Find the posterior probability for every distinct prior

Brute Force MAP Hypothesis

3. Choose the posterior with highest h_{MAP} value

$$\begin{aligned} \theta_{MAP} &= \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n) \\ h_{MAP} &= \arg \max_{h \in H} P(h | D) \\ &= \arg \max_{h \in H} \frac{P(D | h) P(h)}{P(D)} \\ &= \arg \max_{h \in H} P(D | h) P(h) \end{aligned}$$

$$\rightarrow \frac{[P(D | h_1) \cdot P(h_1)]}{P(D)} \leftrightarrow \frac{[P(D | h_2) P(h_2)]}{P(D)}$$

Maximum a Posteriori (MAP) Estimator of θ is the value of θ that maximizes the posterior distribution of θ .

Best hypothesis $\approx$ most probable hypothesis

Maximum A Posteriori Estimation (MAP)

1. Find the prior probability
2. Derive the posterior probability *Bayes.*
3. Differentiate posterior w.r.t. (each) θ
=
4. Solve

Maximum a Posteriori (MAP) Estimator of θ is the value of θ that maximizes the posterior distribution of θ .

Best hypothesis $\approx$ most probable hypothesis

Example :

$P(0.5)$

1. Example on MAP algorithm:

Let X be continuous random variable with probability density function $P(X)$ given by:

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Given another distribution $p(Y|X=x) = x(1-x)^{y-1}$ Find MAP estimate of X given $Y=3$

①

$$h_{\text{MAP}} = P(X | Y=3)$$

$$= P(Y=3 | X) * P(X)$$

$$= x(1-x)^{3-1} * 2x$$

②

To find the parameter x , differentiate the function & equate to zero.

③

$$\frac{d(P(X|Y=3))}{dx} = 0$$

$$\frac{d(x(1-x)^2 * 2x)}{dx} = 0$$

$$\frac{d(2x^2 - 4x^3 + 2x^4)}{dx} = 0$$

$$4x - 12x^2 + 8x^3 = 0$$

$$x = \{0, 0.5, 1\}$$

$$P(\underline{x} | \underline{Y=3}) = \frac{P(Y=3/x) \cdot P(x)}{P(Y=3)}$$

Example :

1. Example on MAP algorithm:

Let X be continuous random variable with probability density function $P(X)$ given by:

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Given another distribution $p(Y|X = x) = x(1 - x)^{y-1}$ Find MAP estimate of X given $Y=3$

$$\begin{aligned} h_{\text{MAP}} &= P(X | Y=3) \\ &= P(Y=3 | X) * P(X) \\ &= x(1-x)^{y-1} * 2x \end{aligned}$$

$$x = \{0, 0.5, 1\}$$

$$P(X|Y=3) = \{0, 0.125, 0\}$$

$$x = 0.5$$

Most Probable Classification of New Instances

- So far we've sought the most probable hypothesis given the data D (i.e., h_{MAP})
- Given new instance x , what is its most probable classification?**
 - $h_{\text{MAP}}(x)$ is not the most probable classification!
 - What's most probable classification of x ?

Consider:

Three possible hypotheses:

$$P(h_1|D) = .4, P(h_2|D) = .3, P(h_3|D) = .3$$

Given new instance x , classification given by above 3 hypotheses is

$$h_1(x) = +, h_2(x) = -, h_3(x) = -$$

1 0 0

$$P(+)+P(-) \neq 1$$

$$P(\Theta|h_1) = 1 \quad \text{and} \quad P(\Theta|h_1) = 0$$

May be the classification for X is $+$

ML setting

- Bayesian Analysis
 - start with some belief about the system, called a prior.
 - Then we obtain some data and use it to update our belief.
 - The outcome is called a posterior.
 - Should we obtain even more data, the old posterior becomes a new prior and the cycle repeats.
 - $P(h | D)$ a posterior determines the class label
 - MLE and MAP are the same if the prior is uniform
 - This forms the basis for Naïve Bayes classifier → More on this will be discussed in later part of this module

belief → evidence → update belief
↓
posterior

• Bayes' Optimal Classifier

- The most probable classification of the new instance is obtained by combining **the predictions of all hypotheses, weighted by their posterior probabilities.**
- v_j from some set V , then the probability $P(v_j | D)$ that the correct classification for the new instance is v_j is:
$$P(v_j | D) = \sum_{h_i \in H} P(v_j | h_i) P(h_i | D)$$
- The optimal classification of the new instance is the value v_j for which $P(v_j | D)$ is maximum

$$\underline{\underline{P(v_j | D) =}}$$

Example 1 : Bayes Optimal Classifier

best possible classification performance

Bayes optimal classification:

$$\arg \max_{v_j \in V} \sum_{h_i \in H} P(v_j | h_i) P(h_i | D)$$

• Example:

$$P(h_1 | D) = .4, P(-|h_1) = 0, P(+|h_1) = 1$$

$$P(h_2 | D) = .3, P(-|h_2) = 1, P(+|h_2) = 0$$

$$P(h_3 | D) = .3, P(-|h_3) = 1, P(+|h_3) = 0$$

therefore

$$\sum_{h_i \in H} P(+|h_i) P(h_i | D) = .4$$

$$\sum_{h_i \in H} P(-|h_i) P(h_i | D) = .6$$

$$\arg \max_{v_j \in V} \sum_{h_i \in H} P(v_j | h_i) P(h_i | D) = -$$

$$\begin{aligned}
 & \downarrow 1 \times 0.4 + \\
 & P(+|h_1) \cdot P(h_1 | D) + \\
 & P(+|h_2) P(h_2 | D) + \\
 & P(+|h_3) P(h_3 | D) \\
 & \quad \quad \quad = 0.4
 \end{aligned}$$

$$P(-|D) = 0.6$$

Gibbs Classifier

- Bayes optimal classifier provides best result, but can be expensive if many hypotheses.
- Gibbs algorithm:
 - Choose one hypothesis at random, according to posterior prob. Distribution over h , $P(h|D)$
 - Use this h to classify new instance

- Surprising fact: under certain conditions, the expected misclassification error for the Gibbs algorithm is at most twice the expected error of the Bayes optimal classifier

$$E[\text{error}_{\text{Gibbs}}] \leq 2E[\text{error}_{\text{BayesOptional}}]$$

- Suppose correct, uniform prior distribution over H , then
 - Pick any hypothesis from *Version space*, with uniform probability
 - Its expected error no worse than twice Bayes optimal

Parameter Estimation in ML - Summary

Assumption: Data are__

- IID samples: $X_1 \dots X_n$
where all X_i are independent and have the same distribution.
- Either same PMF (discrete) or same PDF (continuous)
- $f(X | \theta)$
Likelihood of different values of X depends on the values of our parameters θ
 $f(\cdot)$ is either PDF or PMF

Parameters : $(\theta_0, \theta_1, \theta_2)$

Find $\theta = (\theta_0, \theta_1, \theta_2)$ from the data $\mathbf{X}_i$

Maximum Likelihood Estimation (MLE)

select that parameters θ that make the observed data the most likely

$$f(X_1, X_2, \dots, X_n | \theta)$$

Maximum A Posteriori (MAP)

choose the parameters θ that is the most likely, given the data

$$f(\theta | X_1, X_2, \dots, X_n)$$

If the sample is large, MLE will yield an excellent estimator of θ

When no prior information is available, all hypothesis are equally likely i.e. $p(h_i) = p(h_j)$

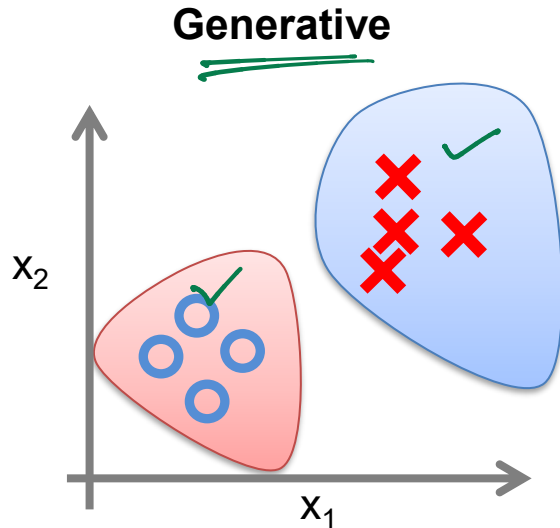
Naïve Bayes Classifier

Refresh from Pre-Mid Term



Decision Theory: Interpretation

Model Building



$$P(Y | X_1 X_2 \dots X_n) = \frac{P(X_1 X_2 \dots X_d | Y) P(Y)}{P(X_1 X_2 \dots X_d)}$$

Known as generative models, because by sampling from them it is possible to generate synthetic data points in the input space.

Eg., Gaussians, **Naïve Bayes**, Mixtures of multinomials, **Mixtures of Gaussians**, Bayesian networks

Handwritten notes: (1) All features are IID, (2) Prior.

Likelihood: $P(x|c)$, Class Prior Probability: $P(c)$

$$P(c|x) = \frac{P(x|c)P(c)}{P(x)}$$

Posterior Probability: $P(c|x)$, Predictor Prior Probability: $P(c)$

$$P(c | X) = P(x_1 | c) \times P(x_2 | c) \times \dots \times P(x_n | c) \times P(c)$$

Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	Normal	Breeze	Same	Yes
Sunny	Hot	Normal	Breeze	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Rainy	Warm	Normal	Breeze	Same	Yes

Naïve Bayes Classifier - Applications

innovate

achieve

lead

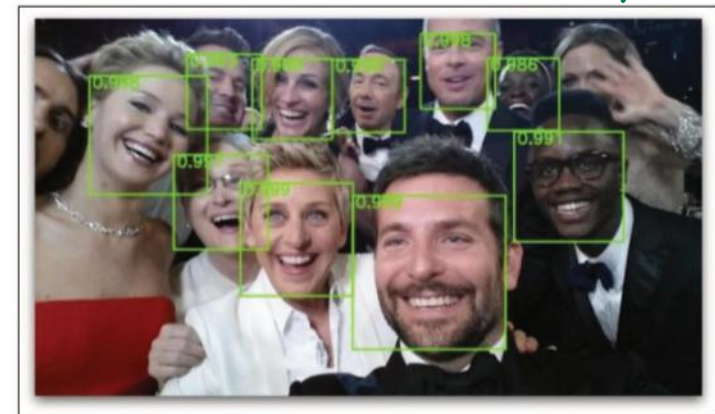
Categorizing News

	BUSINESS & ECONOMY Paying service charge at hotels not mandatory
	TECHNOLOGY & SCIENCE The 'dangers' of being admin of a WhatsApp group
	ENTERTAINMENT This actor stars in Raabta. Guess who?
	IPL 2017 Preview: Bullish KKR face depleted Lions
	INDIA Why is Aadhaar mandatory for PAN? SC asks Centre

Email Spam Detection



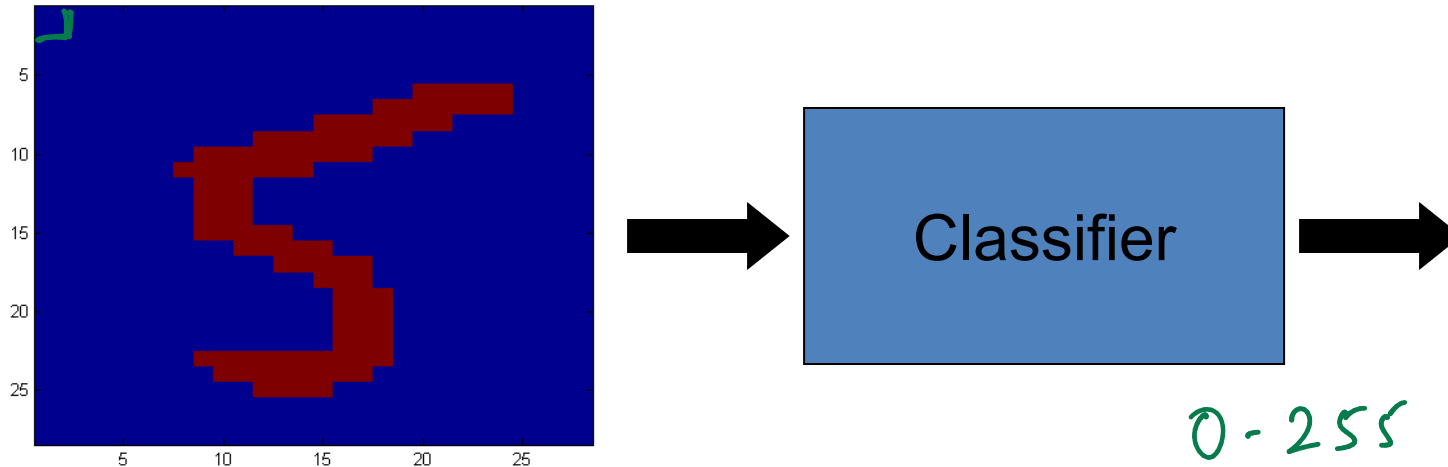
Face Recognition



Sentiment Analysis



Example: Digit Recognition



- $X_1, \dots, X_n \in \{0, 1\}$ (Black vs. White pixels)
- $Y \in \{5, 6\}$ (predict whether a digit is a 5 or a 6)

Generative models for classification



$$p(y|x)$$

$$\underline{p(y|x)} = \left[\frac{p(x|y)p(y)}{\cancel{p(x)}} \right]$$

- For binary classification the denominator is given by

$$p(x) = p(x|y = 1)p(y = 1) + p(x|y = 0)p(y = 0)$$

- if we're calculating $p(y|x)$ in order to make a prediction, then we don't actually need to calculate the denominator, since

$$\begin{aligned} \arg \max_y p(y|x) &= \arg \max_y \frac{p(x|y)p(y)}{p(x)} \\ &= \arg \max_y \underline{p(x|y)p(y)}. \end{aligned}$$

$$\underline{p(x|y)p(y)}$$

The Bayes Classifier

$$X = \{x_1 \dots x_n\}$$

$$y: 5 \quad y: 6$$

$$\begin{aligned} \underline{P(Y = 5|X_1, \dots, X_n)} &= \frac{P(X_1, \dots, X_n|Y = 5)P(Y = 5)}{P(X_1, \dots, X_n|Y = 5)P(Y = 5) + P(X_1, \dots, X_n|Y = 6)P(Y = 6)} \rightarrow \\ \underline{P(Y = 6|X_1, \dots, X_n)} &= \frac{P(X_1, \dots, X_n|Y = 6)P(Y = 6)}{P(X_1, \dots, X_n|Y = 5)P(Y = 5) + P(X_1, \dots, X_n|Y = 6)P(Y = 6)} \leftarrow \end{aligned}$$

- To classify, we'll simply compute these two probabilities and predict based on which one is greater

Naïve Bayes conditional Independence assumption

- Naïve Bayes assumes X_i are conditionally independent given Y

$$P(X_1|X_2, Y) = P(X_1|Y)$$

- Assumption:**

$$P(X_1, \dots, X_n|Y) = \prod_{j=1}^n P(X_j|Y)$$

i.e., X_i and X_j are conditionally independent given Y for $i \neq j$

Naïve Bayes classifier: Prediction

Goal of learning $P(Y|X)$ where $X = \langle X_1, \dots, X_n \rangle$

- Bayes rule:

$$P(Y = y_k | X_1, \dots, X_n) = \frac{P(Y = y_k)P(X_1, \dots, X_n | Y = y_k)}{\sum_j P(Y = y_j)P(X_1, \dots, X_n | Y = y_j)}$$

- Assume conditional independence among X_i 's:

$$P(Y = y_k | X_1, \dots, X_n) = \frac{P(Y = y_k) \prod_i P(X_i | Y = y_k)}{\sum_j P(Y = y_j) \prod_i P(X_i | Y = y_j)}$$

- Classify New Instance(x) : Pick the most probable (MAP) Y for

$$\hat{Y} \leftarrow \underset{y_k}{\operatorname{argmax}} P(Y = y_k) \prod_i P(X_i | Y = y_k)$$

Prior

Likelihood

Example: Play Tennis

$$P(c | x) = \frac{P(x | c)P(c)}{P(x)}$$

Likelihood: $P(x | c)$
 Class Prior Probability: $P(c)$
 Posterior Probability: $P(c | x)$
 Predictor Prior Probability: $P(x)$

$P(X|Y) \sim \text{Multinom}(\pi, n) \rightarrow \text{Multinomial NB } (X_i - \text{multinomial})$

$P(Y) \sim \text{Ber}(p)$

$$P(c | X) = P(x_1 | c) \times P(x_2 | c) \times \dots \times P(x_n | c) \times P(c)$$

$$\hat{Y} \leftarrow \underset{y_k}{\operatorname{argmax}} P(Y = y_k) \prod_i P(X_i | Y = y_k)$$

$P(\text{Yes}) = 3/7$
 $P(\text{No}) = 4/7$

Tr

Test

Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	Normal	Breeze	Same	Yes
Sunny	Hot	Normal	Breeze	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Rainy	Warm	Normal	Breeze	Same	Yes

Lazy

$$\mathcal{X} = \{x_1 \dots x_n\}$$

Example: Play Tennis – Learning Phase

Look up tables

Maximum likelihood estimates (MLE's):

$$\hat{\theta}_{ijk} = \hat{P}(X_i = x_{ij} | Y = y_k) = \frac{\#D\{X_i = x_{ij} \wedge Y = y_k\}}{\#D\{Y = y_k\}} \quad \hat{\pi}_k = \hat{P}(Y = y_k) = \frac{\#D\{Y = y_k\}}{|D|}$$

Number of items in dataset D for which $Y=y_k$

<u>Sky</u>	Play=Yes	Play=No
Sunny	3/3	2/4
Rainy	0/3	2/4

<u>Wind</u>	Play=Yes	Play=No
Strong	2/3	3/4
Breeze	1/3	1/4

<u>AirTemp</u>	Play=Yes	Play=No
Hot	0/3	1/4
Warm	3/3	1/4
Cold	0/3	2/4

<u>Forecast</u>	Play=Yes	Play=No
Same	2/3	2/4
Change	1/3	2/4

<u>Humidity</u>	Play=Yes	Play=No
High	1/3	3/4
Normal	2/3	1/4

Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	Normal	Breeze	Same	Yes
Sunny	Hot	Normal	Breeze	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Rainy	Warm	Normal	Breeze	Same	Yes



Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Sunny	Warm	Normal	Strong	Same	Yes ✓
Sunny	Warm	High	Strong	Same	No —
Rainy	Cold	High	Strong	Change	No —
Sunny	Warm	Normal	Breeze	Same	Yes ✓
Sunny	Hot	Normal	Breeze	Same	No —
Rainy	Cold	High	Strong	Change	No —
Sunny	Warm	High	Strong	Change	Yes ✓
Sunny	Warm	Normal	Strong	Change	???
Rainy	Warm	Normal	Breeze	Same	???

$P(S/Y_{es})$ $P(S/N_{o})$
 ↓

Sky	Y	N
Sunny	3/3	2/4
Rainy	0/3	2/4

 $P(R/Y_{es})$ ↑

Conditional
Prob.
Tables

→

AirT	Y	N
Hot	0/3	1/4
Warm	3/3	1/4
Cold	0/3	2/4



Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	Normal	Breeze	Same	Yes
Sunny	Hot	Normal	Breeze	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Sunny	Warm	Normal	Strong	Change	???
Rainy	Warm	Normal	Breeze	Same	???

① $P(\text{Yes} | x)$

② $P(\text{No} | x)$

} max.
 $\frac{3}{7} = P(\text{Yes})$
 $\frac{4}{7} = P(\text{No})$



$$\textcircled{1} P(\text{Y} | x) = P(\text{Sunny} | \text{Y}) \cdot P(\text{Warm} | \text{Y}) \cdot P(\text{Normal} | \text{Y}) \cdot P(\text{Strong} | \text{Y}) \cdot P(\text{Change} | \text{Y}) * P(\text{Yes})$$

$$P(\text{N} | x) \cdot P(\text{ } | \text{N}) \cdot P(\text{ } | \text{N}) \cdot \dots \cdot x P(\text{No})$$

Example: Play Tennis - Testing Phase

$$P(\text{Enjoy}=\text{Yes} \mid X) = P(X \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Enjoy}=\text{Yes}) / P(X)$$

$$= P(X \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Enjoy}=\text{Yes})$$

$$= P(X \mid \text{Enjoy}=\text{Yes}) \cdot (3/7)$$

$$= P(\text{Sunny} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Warm} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Normal} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Strong} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Change} \mid \text{Enjoy}=\text{Yes}) \cdot (3/7)$$

$$= (3/3) \cdot (3/3) \cdot (2/3) \cdot (2/3) \cdot (1/3) \cdot (3/7)$$

$$= 0.0635$$

$$\hat{\pi}_k = \hat{P}(Y = y_k) = \frac{\#D\{Y = y_k\}}{|D|}$$

$P(\text{Enjoy}=\text{Yes} \mid X) > P(\text{Enjoy}=\text{No} \mid X) \rightarrow \text{EnjoySport} = \text{Yes}$

$$P(\text{Enjoy}=\text{No} \mid X) = P(X \mid \text{Enjoy}=\text{No}) \cdot P(\text{Enjoy}=\text{No}) / P(X)$$

$$= P(X \mid \text{Enjoy}=\text{No}) \cdot P(\text{Enjoy}=\text{No})$$

$$= P(X \mid \text{Enjoy}=\text{No}) \cdot (4/7)$$

$$= P(\text{Sunny} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Warm} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Normal} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Strong} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Change} \mid \text{Enjoy}=\text{No}) \cdot (4/7)$$

$$= (2/4) \cdot (1/4) \cdot (1/4) \cdot (3/4) \cdot (2/4) \cdot (4/7)$$

$$= 0.006696$$

MAP rule

$$Y^{new} \leftarrow \arg \max_{y_k} P(Y = y_k) \prod_i P(X_i^{new} \mid Y = y_k)$$

$$Y^{new} \leftarrow \arg \max_{y_k} \pi_k \prod_i \theta_{ijk}$$

Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	No
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Sunny	Warm	Normal	Breeze	Same	Yes
Sunny	Hot	Normal	Breeze	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Sunny	Warm	Normal	Strong	Change	???
Rainy	Warm	Normal	Breeze	Same	???

Naïve Bayes Algorithm – discrete X_i

- Train Naïve Bayes (examples) for each* value y_k

estimate $\pi_k \equiv P(Y = y_k)$

for each* value x_{ij} of each attribute X_i

estimate $\theta_{ijk} \equiv P(X_i = x_{ij} | Y = y_k)$

- Classify (X^{new})

$$Y^{new} \leftarrow \arg \max_{y_k} P(Y = y_k) \prod_i P(X_i^{new} | Y = y_k)$$

$$Y^{new} \leftarrow \arg \max_{y_k} \pi_k \prod_i \theta_{ijk}$$

Example: Play Tennis

$$\begin{aligned}
 P(\text{Enjoy}=\text{Yes} \mid X) &= P(X \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Enjoy}=\text{Yes}) / P(X) \\
 &= P(X \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Enjoy}=\text{Yes}) \\
 &= P(X \mid \text{Enjoy}=\text{Yes}) \cdot (3/7) \\
 &= P(\text{Rainy} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Warm} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Normal} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Breeze} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Same} \mid \text{Enjoy}=\text{Yes}) \cdot (3/7) \\
 &= (0/3) \cdot (3/3) \cdot (2/3) \cdot (1/3) \cdot (2/3) \cdot (3/7)
 \end{aligned}$$

$$\begin{aligned}
 P(Y/X) &= \frac{0}{3} \times \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 P(\text{Enjoy}=\text{No} \mid X) &= P(X \mid \text{Enjoy}=\text{No}) \cdot P(\text{Enjoy}=\text{No}) / P(X) \\
 &= P(X \mid \text{Enjoy}=\text{No}) \cdot P(\text{Enjoy}=\text{No}) \\
 &= P(X \mid \text{Enjoy}=\text{No}) \cdot (4/7) \\
 &= P(\text{Rainy} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Warm} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Normal} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Breeze} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Same} \mid \text{Enjoy}=\text{No}) \cdot (4/7) \\
 &= (2/4) \cdot (1/4) \cdot (1/4) \cdot (1/4) \cdot (2/4) \cdot (4/7)
 \end{aligned}$$

Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
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Sunny	Warm	Normal	Breeze	Same	Yes
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Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Rainy	Warm	Normal	Breeze	Same	???

Laplace Smoothing



Smoothing

If one of the conditional probabilities is zero, then the entire expression becomes zero

- Technique for smoothing categorical data.
- A small-sample correction, or **pseudo-count**, will be incorporated in every probability estimate.
- No probability will be zero.

Probability estimation:

Original : $P(A_i | C) = \frac{N_{ic}}{N_c}$ ✓

Laplace: $P(A_i | C) = \frac{N_{ic} + 1}{N_c + d}$ ✓ +1 ✓ +1 ✓

m - estimate: $P(A_i | C) = \frac{N_{ic} + mp}{N_c + m}$ ✓ ✓

Bayesian approach

→ d : number of distinct values of an attribute

N_c : number of instances in the class

N_{ic} : number of instances having attribute value A_i in class c

p : prior probability of the class

m : constant called the **equivalent sample size**, which determines how heavily to weight p relative to the observed data

Example: Play Tennis

$$P(\text{Enjoy}=\text{Yes} \mid X) = P(X \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Enjoy}=\text{Yes}) / P(X)$$

$$= P(X \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Enjoy}=\text{Yes})$$

$$= P(X \mid \text{Enjoy}=\text{Yes}) \cdot (3/7)$$

$$= P(\text{Rainy} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Warm} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Normal} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Breeze} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Same} \mid \text{Enjoy}=\text{Yes}) \cdot (3/7)$$

$$= (0+1/3+2) \cdot (3+1/3+3) \cdot (2+1/3+2) \cdot (1+1/3+2) \cdot (2+1/3+2) \cdot (3/7)$$

$$\hat{\theta}_i = \frac{x_i + \alpha}{N + \alpha d} \quad (i = 1, \dots, d)$$

For Laplace smoothing $\alpha = 1$

all attribute / all class

$$P(\text{Enjoy}=\text{No} \mid X) = P(X \mid \text{Enjoy}=\text{No}) \cdot P(\text{Enjoy}=\text{No}) / P(X)$$

$$= P(X \mid \text{Enjoy}=\text{No}) \cdot P(\text{Enjoy}=\text{No})$$

$$= P(X \mid \text{Enjoy}=\text{No}) \cdot (4/7)$$

$$= P(\text{Rainy} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Warm} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Normal} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Breeze} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Same} \mid \text{Enjoy}=\text{No}) \cdot (4/7)$$

$$= (2+1/4+2) \cdot (1+1/4+3) \cdot (1+1/4+2) \cdot (1+1/4+2) \cdot (2+1/4+2) \cdot (4/7)$$

$$P(\text{Rainy} \mid Y) = \frac{0 + 1}{3 + 2}$$

$$P(\text{Warm} \mid Y) = \frac{2 + 1}{3 + 3}$$

Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	Normal	Breeze	Same	Yes
Sunny	Hot	Normal	Breeze	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Sunny	Warm	Normal	Strong	Change	????
Rainy	Warm	Normal	Breeze	Same	????

Example: Play Tennis

$$\begin{aligned}
 P(\text{Enjoy}=\text{Yes} \mid X) &= P(X \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Enjoy}=\text{Yes}) / P(X) \\
 &= P(X \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Enjoy}=\text{Yes}) \\
 &= P(X \mid \text{Enjoy}=\text{Yes}) \cdot (3/7) \\
 &= P(\text{Rainy} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Warm} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Normal} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Breeze} \mid \text{Enjoy}=\text{Yes}) \cdot P(\text{Same} \mid \text{Enjoy}=\text{Yes}) \cdot (3/7) \\
 &= (1/5) \cdot (4/6) \cdot (3/5) \cdot (2/5) \cdot (3/5) \cdot (3/7) \\
 &= 0.00822
 \end{aligned}$$

$P(\text{Enjoy}=\text{Yes} \mid X) > P(\text{Enjoy}=\text{No} \mid X) \rightarrow \text{EnjoySport} = \text{Yes}$

$$\begin{aligned}
 P(\text{Enjoy}=\text{No} \mid X) &= P(X \mid \text{Enjoy}=\text{No}) \cdot P(\text{Enjoy}=\text{No}) / P(X) \\
 &= P(X \mid \text{Enjoy}=\text{No}) \cdot P(\text{Enjoy}=\text{No}) \\
 &= P(X \mid \text{Enjoy}=\text{No}) \cdot (4/7) \\
 &= P(\text{Rainy} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Warm} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Normal} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Breeze} \mid \text{Enjoy}=\text{No}) \cdot P(\text{Same} \mid \text{Enjoy}=\text{No}) \cdot (4/7) \\
 &= (3/6) \cdot (2/7) \cdot (2/6) \cdot (2/6) \cdot (3/6) \cdot (4/7) \\
 &= 0.00453
 \end{aligned}$$

Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	Normal	Breeze	Same	Yes
Sunny	Hot	Normal	Breeze	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Sunny	Warm	Normal	Strong	Change	???
Rainy	Warm	Normal	Breeze	Same	???

Naïve Bayes: Continuous Features

- X_i can be continuous

→ Age, Salary, Ht,

Naïve Bayes classifier:

$$Y = \arg \max_y P(Y = y) \prod_i P(X_i | Y = y)$$

Assumption: $P(X_i | Y)$ has a **Gaussian** distribution

Yes No



within each class

The Gaussian Probability Distribution

- It is a continuous distribution with pdf:

μ = mean of distribution

σ^2 = variance of distribution

x is a continuous variable ($-\infty \leq x \leq \infty$)

σ^2

σ SD

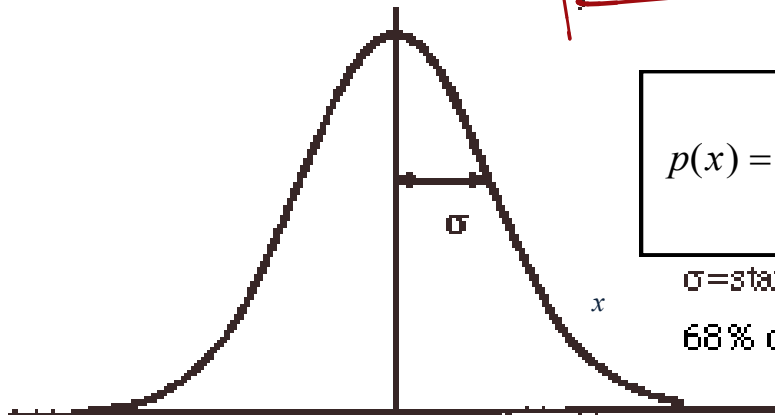
$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$x = 35$

$P(\text{Age}^x | \text{Yes})$

μ
 σ^2 / σ

mode=median=mean = μ



$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \text{ gaussian}$$

σ =standard deviation

68% of area within $\pm 1\sigma$

Continuous Features : learning and prediction

- For each target value Y_k (MLE estimate)

$$P(Y = y_k) \leftarrow \text{No. of instances with } Y_k \text{ class} / \text{No. of Total instances}$$

- For each attribute value X_i estimate $P(X_i | Y = y_k)$

– class conditional mean, variance

Prior

No table

→ for each class.

- Classify New Instance(x)

Pick the most probable (MAP) Y

$$\hat{Y} \leftarrow \underset{y_k}{\operatorname{argmax}} P(Y = y_k) \prod_i P(X_i | Y = y_k)$$

Continuous Features : learning

- $P(X_i|Y)$ is Gaussian
- Training: estimate mean and standard deviation
 - $\mu_i = E[X_i|Y = y]$ ✓
 - $\sigma_i^2 = E[(X_i - \mu_i)^2|Y = y]$ ✓

X_1	X_2	X_3	Y
2	3	1	1
-1.2	2	0.4	1
1.2	0.3	0	0
2.2	1.1	0	1

Continuous Features : learning



$(X_i | Y=y) \rightarrow \mu, \sigma^2$

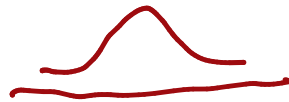
X_1	X_2	X_3	Y
2 ✓	3	1	1
-1.2 ✓	2	0.4	1
1.2	0.3	0	0
2.2 ✓	1.1	0	1

- $\mu_i = E[X_i | Y = y]$
- $\sigma_i^2 = E[(X_i - \mu_i)^2 | Y = y]$

$$\begin{aligned} \checkmark - \mu_1 &= E[X_1 | Y = 1] = \frac{2 + (-1.2) + 2.2}{3} = 1 \\ \checkmark - \sigma_1^2 &= E[(X_1 - \mu_1)^2 | Y = 1] = \frac{(2 - 1)^2 + (-1.2 - 1)^2 + (2.2 - 1)^2}{3} = 2.43 \end{aligned}$$

$$\frac{2 + (-1.2) + 2.2}{3} = 1$$

$$\frac{(2-1)^2 + (-1.2-1)^2 + (2.2-1)^2}{3} = \sum \frac{(x_i - \mu)^2}{n}$$



Example: Play Tennis

$P(X|Y) \sim N(\mu, \sigma^2) \rightarrow$ GaussianNB (X_i – real valued)

$$P(\text{Enjoy}=\text{Yes} | X) = P(X | \text{Enjoy}=\text{Yes}) \cdot P(\text{Enjoy}=\text{Yes}) / P(X)$$

$$= P(X | \text{Enjoy}=\text{Yes}) \cdot P(\text{Enjoy}=\text{Yes})$$

$$= P(X | \text{Enjoy}=\text{Yes}) \cdot (3/7)$$

$$= P(\text{Rainy} | \text{Enjoy}=\text{Yes}) \cdot P(\text{Warm} | \text{Enjoy}=\text{Yes}) \cdot P(60 | \text{Enjoy}=\text{Yes}) \cdot P(\text{Breeze} | \text{Enjoy}=\text{Yes}) \cdot P(\text{Same} | \text{Enjoy}=\text{Yes}) \cdot (3/7)$$

$$= (1/3) \cdot (3/3) \cdot 0.15 \cdot 10^{-95} \cdot (1/3) \cdot (2/3) \cdot (3/7)$$

$$\mu_i = E[X_i | Y = \text{yes}] = 84.33$$

$$\sigma_i^2 = E[(X_i - \mu_i)^2 | Y = \text{yes}] = 1.15$$

$$\mu_i = E[X_i | Y = \text{no}] = 72.5$$

$$\sigma_i^2 = E[(X_i - \mu_i)^2 | Y = \text{no}] = 17.08$$

$$x = 60$$

$$P(\text{Enjoy}=\text{No} | X) = P(X | \text{Enjoy}=\text{No}) \cdot P(\text{Enjoy}=\text{No}) / P(X)$$

$$= P(X | \text{Enjoy}=\text{No}) \cdot P(\text{Enjoy}=\text{No})$$

$$= P(X | \text{Enjoy}=\text{No}) \cdot (4/7)$$

$$= P(\text{Rainy} | \text{Enjoy}=\text{No}) \cdot P(\text{Warm} | \text{Enjoy}=\text{No}) \cdot P(60 | \text{Enjoy}=\text{No}) \cdot P(\text{Breeze} | \text{Enjoy}=\text{No}) \cdot P(\text{Same} | \text{Enjoy}=\text{No}) \cdot (4/7)$$

$$= (2/4) \cdot (1/4) \cdot 0.02 \cdot (1/4) \cdot (2/4) \cdot (4/7)$$

$$P(60 | \text{Yes}) \rightarrow \frac{1}{\sqrt{2\pi(1.15)}} e^{-\frac{(60-84.33)^2}{2(1.15)}}$$

Humidity	Enjoy Sport?	AirTemp	Sky	Wind	Forecast	Enjoy Sport?
85	Yes	Warm	Sunny	Strong	Same	Yes
80	No	Warm	Sunny	Strong	Same	No
70	No	Cold	Rainy	Strong	Change	No
83	Yes	Warm	Rainy	Breeze	Same	Yes
90	No	Hot	Sunny	Breeze	Same	No
50	No	Cold	Rainy	Strong	Change	No
85	Yes	Warm	Sunny	Strong	Change	Yes
60	???	Warm	Rainy	Breeze	Same	???

$$x = 60$$

$$\mu, \sigma$$

Additional Exercises – For Student's Practice

As a part of efforts to improve students' performance in the exams, you have been given the data showing number of study hours spent by students, their gender and their final results as pass or fail. Using this sample dataset, apply Naïve Bayes classification technique, to classify the below test case:

{No of study hours = 3.5, Gender="male"} either as "Pass", or "Fail".

Tip: $P(X|Y) \sim N(\mu, \sigma^2) \rightarrow$ GaussianNB (X_i – real valued)

No of study hours	Gender	Final result
4.5	Male	Pass
7	Female	Pass
2	Male	Fail
4	Female	Fail
2.5	Male	Fail
3	Female	Fail
8.3	Male	Fail
8	Female	Pass
9	Male	Pass

Learning

- ① Prior
- ② table for categorical
- ③ μ, σ^2 for numerical

$$\underline{\underline{P(\text{Yes} | X)}} = \underbrace{\prod P(x_i | \text{Yes})}_{\text{discrete}} \cdot P(\text{Yes})$$

```
graph TD; A["(Product Term)"] --> B["discrete"]; B --> C["count"]; C --> D["table"]; A --> E["cont."]; E --> F["Gaussian"];
```

discrete
↓
count
↓
table

cont.
↓
Gaussian

Additional Exercises – For Student's Practice

$$\begin{aligned}
 P(\text{Pass} | X) &= P(X | \text{Pass}). P(\text{Pass}) / P(X) \\
 &= P(X | \text{Pass}). P(\text{Pass}) \\
 &= P(X | \text{Pass}). (0.44) \\
 &= P(\text{Male} | \text{Pass}). P(3.5 | \text{Pass}). (0.44) \\
 &= (2/4). 0.1056 . (0.44) = 0.0235
 \end{aligned}$$

$$p(X_i = x | Y = y_k) = \frac{1}{\sqrt{2\pi\sigma_{ik}^2}} e^{-\frac{1}{2}\left(\frac{x-\mu_{ik}}{\sigma_{ik}}\right)^2}$$

$$\begin{aligned}
 P(\text{Fail} | X) &= P(X | \text{Fail}). P(\text{Fail}) / P(X) \\
 &= P(X | \text{Fail}). P(\text{Fail}) \\
 &= P(X | \text{Fail}). (0.56) \\
 &= P(\text{Male} | \text{Fail}). P(3.5 | \text{Fail}). (0.56) \\
 &= (3/5). 0.1846 . (0.56) = 0.0615
 \end{aligned}$$

No of study hours	Gender	Final result
4.5	Male	Pass
7	Female	Pass
2	Male	Fail
4	Female	Fail
2.5	Male	Fail
3	Female	Fail
8.3	Male	Fail
8	Female	Pass
9	Male	Pass

Thank you !

Required Reading for completed session :

T1 - Chapter # 8 (Tom M. Mitchell, Machine Learning)

T1 - Chapter # 6 (Tom M. Mitchell, Machine Learning)

R1 – Chapter # 3 (Christopher M. Bhisop, Pattern Recognition & Machine Learning)

& Refresh your MFDS & ISM parallel course basics